

SecValid AI

End-to-End Security Workflow

Architectural flow, 13-category hybrid analysis engine, and AI-powered remediation pipeline documentation.

REVISION: 2026.02.10

STATUS: CLASSIFIED - SECURE

1. The 4-Stage Analysis Pipeline

Stage 1: Discovery & Passive Reconnaissance

When a user submits a URL, the engine initiates a 'clean' fetch. Captures full page source, including inline scripts and meta tags. Probes HTTP response headers for security flags (CSP, HSTS). Automatically checks for infrastructure files like /robots.txt and /ads.txt.

Stage 2: 13-Category Hybrid Engine

SecValid AI 2.0 uses a unique hybrid engine that runs 13 specialized categories in parallel. Combining Deterministic Static Analysis (Regex) for speed with Generative AI Enrichment (Gemini 2.0) for context-aware logic auditing.

Stage 3: Intelligence Fallback Layer

To ensure the scanner never fails, we implement a 3-Tier AI Reliability Strategy: Gemini 1.5 Flash (Speed), Gemini 1.5 Pro (Precision), and Static Recovery (Baseline).

Stage 4: Synthesis & Remediation

The 'brain' aggregates all results into a structured JSON model. Includes Weighted Risk Scoring and exact fixCode blocks for developers to copy-paste directly.

2. The 13-Category Security Checklist

Core Logic Categories

1. Injection Attacks: SQLi, NoSQLi, Command Injection.
2. Broken Authentication: Session flags, Weak MFA.
3. XSS: Reflected, Stored, and DOM-based.
4. Security Misconfig: Verbose errors, Test routes.
5. Sensitive Data: PII exposure, weak encryption.
6. OWASP Top 10: IDOR, SSRF, XXE.
7. Supply Chain: Outdated frameworks.
8. CSRF: Missing tokens, Clickjacking.

Advanced Intelligence Categories

- 9. API Security: CORS, GraphQL, Secrets.
- 10. Compliance: GDPR, HIPAA, PCI-DSS logic.
- 11. Privacy: Tracker detection, Stealth collection.
- 12. Business Logic: Workflow skips, Param tampering.
- 13. Deceptive Design: Phishing, UI Mimicry.

3. LinkedIn Post & Professional Outreach

Key Competitive Advantages

- Hybrid Accuracy: Regex precision meets AI intuition.
- Active Remediation: We don't just find, we fix.
- Privacy First: Local analysis with secure AI routing.
- Zero Configuration: Just enter a URL and hit Analyze.